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CYS103 Linear Algebra

Chapter 2

Matrix Algebra

2.1 Matrix Operations

2.2 The inverse of A Matrix

2.3 Dimension and Rank

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Outline:

2.1 Matrix Operations

2.2 The inverse of A Matrix

2.3 Dimension and Rank

2.1 Matrix Operations

Matrix Notation

Two ways to denote $m \times n$ matrix A:

In terms of the *columns* of A:

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$$

In terms of the *entries* of A:

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix}$$

Diagonal matrix:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Main diagonal entries: $a_{11}, a_{22}, a_{33}, \ldots$

Identity matrix:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2.1 Matrix Operations

Sum and Scalar Multiples

Properties

Let A, B, and C be matrices of the same size, and let r and s be scalars:

a) $A + B = B + A$

b) $(A + B) + C = A + (B + C)$

c) $A + 0 = A$

d) $r(A + B) = rA + rB$

e) $(r + s)A = rA + sA$

f) $r(sA) = (rs)A$

Note:

Example:

Let $A = \begin{bmatrix} 4 & 0 & 5 \\ -1 & 3 & 2 \end{bmatrix}$, $B = \begin{bmatrix} 2 & -3 \\ 0 & 1 \end{bmatrix}$, and $C = \begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 7 \end{bmatrix}$. Find

1) $A + B$

2) $A + C$

3) $A - 2C$

2.1 Matrix Operations

Matrix Multiplication

Definition

If A is an $m \times n$ matrix and B is an $n \times p$ matrix with columns $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_p$. Then the product AB is the $m \times p$ matrix whose columns are $A\mathbf{b}_1, A\mathbf{b}_2, \ldots, A\mathbf{b}_p$.

$$AB = A\begin{bmatrix}\mathbf{b}_1 & \mathbf{b}_2 & \ldots & \mathbf{b}_p\end{bmatrix}$$
$$= \begin{bmatrix}A\mathbf{b}_1 & A\mathbf{b}_2 & \ldots & A\mathbf{b}_p\end{bmatrix}$$

Note:

If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.

Example:

Let $A = \begin{bmatrix}2 & 3\\ 1 & -5\end{bmatrix}$, $B = \begin{bmatrix}4 & 3 & 6\\ 1 & -2 & 3\end{bmatrix}$, and $C = \begin{bmatrix}1 & 1 & 1\\ 2 & 3 & 1\\ 0 & 1 & 1\end{bmatrix}$. Find

1) AB

2) AC

2.1 Matrix Operations



The Properties of Matrix Multiplication

Properties

Let A be an $m \times n$ matrix, B and C have sizes for which the indicated **sums** and **products** are defined.

a) $A(BC) = (AB)C$ ---------- associative law of multiplication

b) $A(B + C) = AB + AC$ ---------- left distributive law

c) $(B + C)A = BA + CA$ ---------- right distributive law

d) $r(AB) = (rA)B = A(rB)$, for any scalar r.

e) $I_m A = A = AI_n$ ---------- identity for matrix multiplication

Note:

- Powers of a matrix

  A^k denotes the product of k copies of A: $A^k = A \cdots A$, where $k \in \mathbb{Z}$.

2.1 Matrix Operations

The Transpose of A Matrix

Definition

Given an $m \times n$ matrix A, the transpose of A, denoted by A^T, is the $n \times m$ matrix.

Theorem

Let A and B denote matrices whose sizes are same,

a) $(A^T)^T = A$

b) $(A + B)^T = A^T + B^T$

c) $(rA)^T = rA^T$, for any scalar r

d) $(AB)^T = B^T A^T$

Example:

Let $B = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}$. Find

1) B^T

2) $(2B)^T$

2.2 The Inverse of A Matrix

The Inverse of 2x2 Matrix

Theorem

Let an 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is **invertible** and the inverse of A is:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

If $ad - bc = 0$, then A is **not invertible**.

Chapter 3
Determinant:
$$\det A = ad - bc$$

Note:

- Singular matrix: a matrix is not invertible.
- Nonsingular matrix: a matrix is invertible.

Note:

$$AA^{-1} = A^{-1}A = I_n$$

2.2 The Inverse of A Matrix

Example:

Find the inverse of $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$.

2.2 The Inverse of A Matrix

Example:

Use the inverse of matrix A to solve the following system.

$$\begin{aligned} 3x_1 &+ 4x_2 = 3 \\ 5x_1 &+ 6x_2 = 7 \end{aligned}$$

Note:

If A is an invertible $n \times n$ matrix, then the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution ($\mathbf{x} = A^{-1}\mathbf{b}$).

2.2 The Inverse of A Matrix

The Inverse of $n \times n$ Matrix: Algorihm

❑ **Algorithm** for finding A^{-1}, where A is an $n \times n$ matrix and $n \geq 1$:

1) Row reduced the augmented matrix: $[A \quad I]$.

2) $[A \quad I] = [I \quad A^{-1}]$,

where A is row equivalent to I. Otherwise, A does NOT have an inverse.

Example:

Find the inverse of the matrix $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{bmatrix}$, if it exists.

2.2 The Inverse of A Matrix

The Characterisations of Invertible Matrices

The Invertible Matrix Theorem

Let A be a square $n \times n$ matrix,

1) A is invertible matrix.

2) A is row equivalent to the $n \times n$ identity matrix, I_n.

3) A has n pivot positions.

4) The homogeneous equation $A\mathbf{x} = 0$ has only the trivial solution.

5) The matrix equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$ in $\mathbb{R}^n$ (unique solution $\mathbf{x} = A^{-1}\mathbf{b}$).

6) A^T is an invertible matrix.

2.2 The Inverse of A Matrix

Example:

Use the Invertible Matrix Theorem to determine the following matrix A is invertible.

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 3 & 1 & -2 \\ -5 & -1 & 9 \end{bmatrix}$$

2.3 Dimension and Rank

Dimension and Rank of a Subspace

❑ Subspaces of $\mathbb{R}^n$

W is a set of vector in $\mathbb{R}^n$.

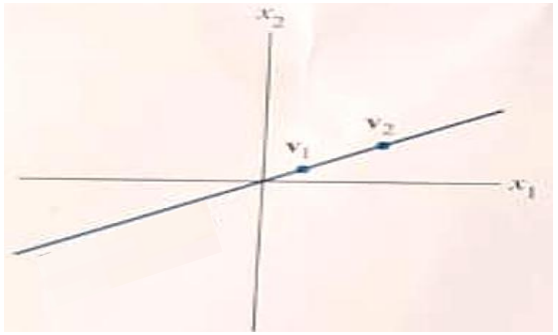
A subspace in $\mathbb{R}^n$ is a subset of $\mathbb{R}^n$ that is itself under the vector addition and scalar multiplication.

Definition

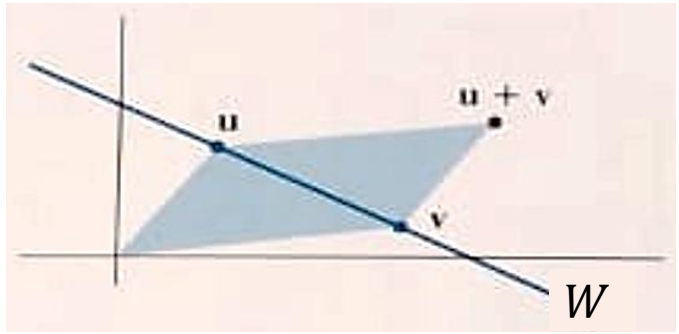
A **subspace** of $\mathbb{R}^n$ is any set W in $\mathbb{R}^n$ that consists of the following properties:

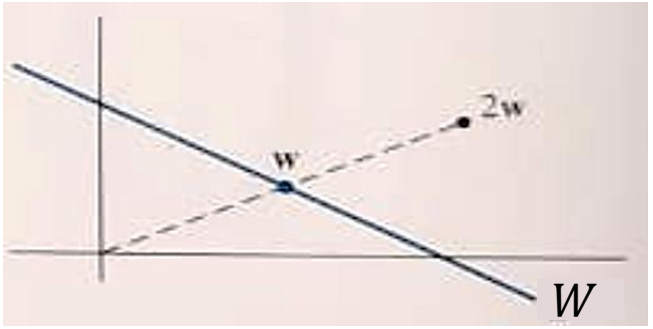
1) The **zero vector** if in W.
2) For each $\mathbf{u}$ and $\mathbf{v}$ in W, the sum $\mathbf{u} + \mathbf{v}$ is in W.
3) For each $\mathbf{u}$ in W and a scalar c, the vector $c\mathbf{u}$ is in W.

Example:

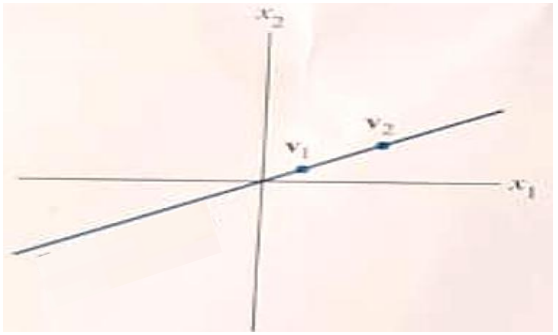


- $\mathbf{v}_1$ is not zero
- $\mathbf{v}_2$ is a multiple of $\mathbf{v}_1$
- So, a line through origin is a subspace.

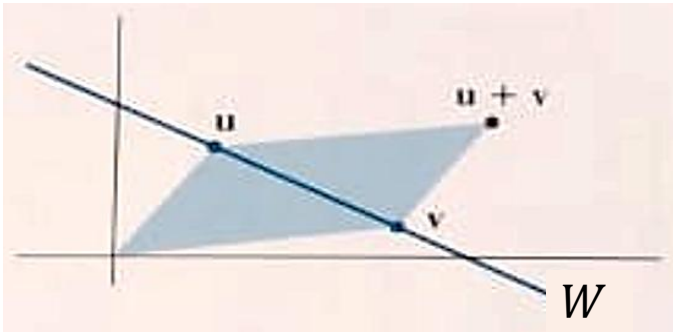


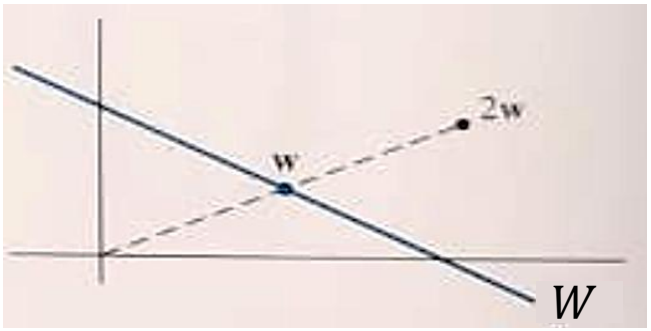


- W is not under addition or scalar multiplication.
- So, a line W is not a subspace as it is not through the origin (it does not consists of the origin).



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2.3 Dimension and Rank

❑ Subspaces of $\mathbb{R}^n$

Example:

$$W = \{(x, y) \in \mathbb{R}^2 \mid y = 2x\}$$

1. Contains the zero vector

   $\mathbf{0} \in W$

2. Closed under vector addition

   If $\mathbf{u}, \mathbf{v} \in W$, then

   $\mathbf{u} + \mathbf{v} \in W$

3. Closed under scalar multiplication

   If $\mathbf{u} \in W$ and $c \in \mathbb{R}$, then

   $c\mathbf{u} \in W$

2.3 Dimension and Rank

Dimension and Rank of a Subspace

Definition

The **dimension** of a nonzero subspace W, denoted by $\dim W$, is the number of vectors in any basis for W.

Dimension of the null space of A ($\text{Nul } A$): the number of free variables in $A\mathbf{x} = \mathbf{b}$.

Definition

The **rank** of a matrix A, denoted by $\text{rank } A$, is the number of pivot columns in A.

Dimension of the column space of A ($\text{Col } A$)

The Rank Theorem

For a matrix A with n columns,

$$\text{rank } A + \dim \text{Nul } A = n$$

A basis of a vector space W is a set of vectors that:

a) Span the space (can generate every vector in W)

b) Are linearly independent (the vectors not multiple of each other).

2.3 Dimension and Rank

Example:

Find the rank of the matrix $A = \begin{bmatrix} 2 & 5 & -3 & -4 & 8 \\ 4 & 7 & -4 & -3 & 9 \\ 6 & 9 & -5 & 2 & 4 \\ 0 & -9 & 6 & 5 & -6 \end{bmatrix}$

The End.